

COURSE TITLE : INSTRUMENTATION AND PROCESS CONTROL
COURSE CODE : 6071
COURSE CATEGORY : A
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Measuring Instruments for temperature, pressure, flow and level.	19
2	Measuring Instruments for specific gravity, Viscosity & Humidity	18
3	Composition analysis.	19
4	Automatic process control	19
TOTAL		75

COURSE OUTCOME :

SL.NO.	SUB	STUDENT WILL BE ABLE TO
1	1	Understand the importance of instrumentation in chemical process industries
	2	Understand the working of instruments used for measuring temperature, pressure, flow & level
2	3	Understand the working of instruments used for measuring viscosity, specific gravity & humidity & pH
3	4	Understand the methods of composition analysis
4	5	Understand process control variables, controllers & control valves
	6	Understand computerized process control system

SPECIFIC OUTCOMES

MODULE - I

Measuring instrument for Temperature, Pressure, Flow and Level measurement

1.1.0 Understand the importance of instrumentation in chemical process industries

- 1.1.1 List important process variables in chemical industry
- 1.1.2 Static and dynamic characteristics of instruments
- 1.1.3 List the basic elements of measuring instrument
- 1.1.4 Draw a sketch of measuring instrument and indicate the basic elements and explain the functions of these elements
- 1.1.5 List the temperature measuring instruments
- 1.1.6 Explain the working of pressure spring thermometers
- 1.1.7 Explain the working of pneumatic balance pressure thermometer
- 1.1.8 Explain the working of vapour pressure thermometers
- 1.1.9 Explain the working principles of thermocouples – SEEBACK, THOMSON AND PETLIER effect

- 1.1.10 List the commonly used thermocouple wires with their temperature range of measurement
- 1.1.11 List the advantages and disadvantages of thermocouples
- 1.1.12 Explain the principle and working of radiation pyrometers
- 1.1.13 Explain the principle and working of optical pyrometers
- 1.2.0 Understand the working of pressure measuring instruments**
 - 1.2.1 Explain the working principle of a 'c' type Bourden tube
 - 1.2.2 Explain the following gauges with sketch. (1) McLeod gauge (2) Pirani gauge
 - 1.2.3 Explain the working of Bellows gauge and diaphragm gauge
- 1.3.0 Understand the working of various flow measuring instruments**
 - 1.3.1 Principles involved in various flow meters
 - 1.3.2 Explain the working of gas flow meter
- 1.4.0 Understand the working of various liquid level measuring instruments**
 - 1.4.1 Explain the working of (1) float type (2) Diaphragm box (3) Bubbler system (4) Electronic signalling type (5) Magnetic type

MODULE - II

Measuring Instruments for specific gravity, Viscosity Humidity

- 2.1.0 Understand the working of instruments used for measuring viscosity, specific gravity, humidity.**
 - 2.1.1 Describe with sketches of the following viscosity measuring instruments
(1) Ostwald viscometer (2) Continuous viscometer
 - 2.1.2 Describe with the sketch displacement meter for specific gravity measurement
 - 2.1.3 Describe with sketch hydrometer type density measuring Instrument
 - 2.1.3 List the various humidity measuring instruments
 - 2.1.4 Define – Absolute humidity, specific humidity, relative humidity, wet bulb and dry bulb Temperature, dew point temperature
 - 2.1.5 Explain with sketches of the following instruments
(i) Wet and dry bulb thermometer (ii) Hair hygrometer (iii) Electric hygrometer
(iv) Dew point recorder
- 2.2.0 Understand the significance of pH in process industries**
 - 2.2.1 List the pH scale, with pH of some known liquids
 - 2.2.2 Describe with sketches of glass electrode and calomel electrode
 - 2.2.3 Explain the basic working principles of a pH meter
 - 2.2.4 List various types of sensing devices of pH meter
 - 2.2.5 Explain immersed type sensor for pH meter

MODULE-III

Composition analysis methods

3.1.0 Understand the methods of composition analysis

- 3.1.1 Define spectroscopy
- 3.1.2 Classification of spectroscopy
- 3.1.3 Describe with a sketch of an infrared spectrophotometer
- 3.1.4 Application of infrared spectrophotometer
- 3.1.5 Describe with a sketch an ultraviolet spectrophotometer
- 3.1.6 Application of ultraviolet spectroscopy
- 3.1.7 Describe with a sketch of a flame photometer and lists its application
- 3.1.8 Advantages and drawbacks of flame photometer
- 3.1.9 Describe with a sketch of a Mass spectrometer and lists its application
- 3.1.10 Disadvantages of mass spectroscopy
- 3.1.11 Describe with a sketch of Atomic Absorption spectroscopy and list its application

3.2.0 Understand chromatography and its classification

- 3.2.1 Define chromatography
- 3.2.2 List the advantages of chromatography
- 3.2.3 Explain with sketch of adsorption chromatography
- 3.2.4 Explain with sketch of Liquid column Chromatography
- 3.2.5 Explain Gas Solid Chromatography and Gas Liquid Chromatography with neat sketch
- 3.2.6 Comparison of GSC and GLC
- 3.2.7 Explain with sketch of HPLC

MODULE - IV

Automatic process control

4.1.0 Understand the important Process control variables and terms

- 4.1.1 Define 1) Controlled variable 2) Manipulated variable 3) Load variable 4) Dead time 5) transportation lag 6) Resistance 7) Capacitance
- 4.1.2 Explain the following types of control action
 - 1) Proportional 2) Integral 3) Proportional integral 4) Proportional Derivative 5) PID
- 4.1.3 Explain two position control and its working
- 4.1.4 List various controlling methods
- 4.1.5 Explain Closed loop system or feedback control system with neat fig and example.
- 4.1.6 Explain Open loop system or feed forward control system with neat fig and example
- 4.1.7 Differentiate different types of control actions and how they are selected with neat figure.
- 4.1.8 Describe in detail the temperature control of an air oven with two position control
- 4.1.9 Draw the Instrumentation diagram of the following a) extraction column b) Single effect evaporator c) chemical reactor

4.2.0 Understand Controllers and control valves

- 4.2.1 Explain the important features of controllers
- 4.2.2 Explain working of pressure regulator
- 4.2.3 Explain the working of flapper nozzle pneumatic controllers
- 4.2.4 Differentiate electronic and pneumatic controller

- 4.2.5 Explain the working of various types control valve plugs
- 4.2.6 Define final control element.
- 4.2.7 List various forces acting on the output member of final control element.
- 4.2.8 Explain fluid flow control valve with neat sketch
- 4.2.9 Explain solenoid valve

4.3.0 Understand the computerized control system

- 4.3.1 Explain a micro-processor with neat sketch
- 4.3.2 Define analog and digital signal transmission
- 4.3.3 State the functions of A/D converter and D/A converter
- 4.3.4 Explain DCS with a neat diagram
- 4.3.5 Explain the DCS components
- 4.3.6 State the Application of DCS
- 4.3.7 List advantages and limitations of DCS
- 4.3.8 Explain the following
 - a) Control components of SCADA
 - b) Working of SCADA
 - c) Comparison of SCADA with DCS
- 4.3.9 Describe the requirements of digital control and information monitoring system
- 4.3.10 Draw the standard symbols of instruments and controllers
- 4.3.11 List the functions of control room instruments

COURSE CONTENT

MODULE – I

Measuring Instruments – Temperature, Pressure Measurements Flow & Level Measurement

Principles of measurement – Instruments for indication, Process variables to be measured in chemical industries- static and dynamic characteristics - elementary principles and description of the instruments used for the measurement of the following;

1. Temperature – filled system thermometers –thermocouples – resistance thermometers – total radiation and optical pyrometers.
2. Pressure and vacuum – manometers – diaphragm gauges, bellow gauges, Bourdon

gauges, absolute pressure gauges, Mc load gauges pirani gauge, piezo resistance and ezo electric transducers.

Different flow meters and level measuring instruments

1. Level measurement: float type, displacement type , diaphragm box, bubbler system, electrical methods, radiation method

MODULE – II

Understand the measuring instrument for viscosity, specific gravity, humidity

Viscosity—Ostwald-continuous viscometer Sp. Gravity- Displacement meter, hydrometer

Humidity

Absolute humidity, specific humidity, relative humidity, wet bulb and dry bulb temperature, dewpoint temperature

Dew point method, wet bulb method, hygrometry, electrical type. pH- significance- scale-liquids, Electrodes- glass, calomel, pH meter- principle-sensing elements-immersed type

MODULE – III

Understand the composition analysis

Methods of compositional analysis- spectroscopy, spectrophotometer-IR-UV-flame photometer-mass spectrophotometer-AAS- advantages, applications of the above spectrophotometers
Chromatographic analysis – adsorption, LCC-GSC-GLC-HPLC- advantages and applications

MODULE – IV

Process Control

Process -control variables-terms types of control action-proportional integral, PI-PD-PID process control system open and closed loop systems – block diagrams of back ward feed, forward feed controls – types of control modes – on-off, proportional, integral, derivative, and their combinations. Comparison of different types of controls- temperature control of air oven instrumentation diagrams- various controlling methods-- feedback control- feed forward control- -, controllers-working of flapper nozzle Pneumatic control-Various control valve plugs-. Final control elements – control valves, actuators, fluid flow control valve

Computerized Control and Instrumentation Diagrams

Microprocessor

Descriptive treatment of the following: -

Analog and digital signal transmission, A/D, D/A converter, analog and digital computer control system – microprocessor, distributed control system –SCADA,Comparison of SCADA with DCS.

REFERENCE

- 1 Dr.A. Sooryanarayana - Outlines of chemical instrumentation & Process control Khanna
- 2 R.K Jain - Mechanical& Industrial Measurements Khanna
- 3 Donald P. Eckman - Industrial Instrumentation CBS, Automatic Process
- 4 Jain& Jain - Engineering Chemistry Danapath Rai